
Predicting University Website Quality from Measurable Page Attributes

A Two-Track Labelling Design that Avoids Circular Evaluation

Course: Machine Learning Laboratory
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Abstract

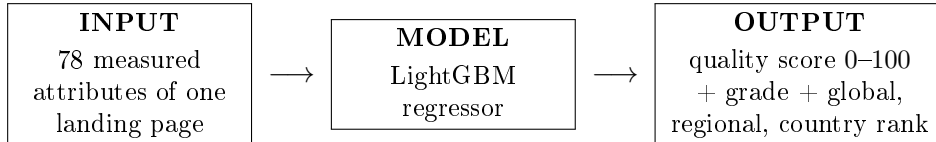
We build a model that reads 78 measurable attributes of a university landing page and returns a 0–100 website-quality score, then use it to rank 1,226 university websites globally. The central difficulty is not modelling but *labelling*: the dataset ships without a target column, and the obvious fix — inventing a weighted formula over the features and training a model to predict it — produces a near-perfect R^2 that measures nothing. We therefore separate the two roles. Track A is an explicit, auditable rubric that serves as the *baseline*; Track B is a set of 900 blind pairwise judgments over 200 anonymised profile cards, converted to a latent score by the Bradley–Terry model, which serves as the *target*. Because Track B is not a closed-form function of the features, $\text{Spearman}(A, B)$ becomes a real empirical result rather than an identity.

Our labelling was performed twice, and the difference between the two passes is the paper’s main methodological finding. The first pass produced a label dominated by *alt-text coverage* — an attribute no site visitor can perceive — because the profile cards advertised it as a headline number, and it was further corrupted by median imputation that described 469 sites with no dated notice at all as having “posted yesterday”. After fixing the missing-value policy and re-framing the judging question around what a prospective applicant actually needs, all 900 pairs were re-judged. Alt-text citations fell from 37.6% to 0%, admission-task content rose from 38.1% to 73.1%, and 16.4% of winners changed — the signature of a genuine criterion change rather than noise. Self-consistency on swapped repeat pairs rose to 96.7%.

On the corrected label, a LightGBM regressor reaches Spearman $\rho = 0.922$ ($R^2 = 0.818$, MAE = 9.0 points) on a 40-university test set held out from every fitting and tuning decision, against $\rho = 0.810$ for the transparent rubric it must beat. Leave-one-region-out validation gives $\rho = 0.844$, confirming the model learned quality rather than geography.

1 Introduction and Problem Statement

1.1 What the system does



Concretely, given a row such as “lists programmes: yes; departments page: yes; admissions policy: no; latest notice: 30 days old; menu: 6 items; contrast 13.3:1; mobile score 75”, the system returns **85.9 / 100, grade A+, global rank 31 of 1226**. Higher is better throughout: the best sites score near 92, the worst near 0.

1.2 Why this is not a trivial regression

A quality score has to come from somewhere. The tempting shortcut is to define $y = 0.5c + 0.3u + 0.2s$ over content, usability and speed features, then train a model to predict y . This reports $R^2 \approx 0.98$ and proves only that a learner can re-derive our own arithmetic. Biyyapu et al. (2023) published exactly this design and reported 98.2% accuracy for predicting a lookup table from its own inputs. Avoiding that circularity is the organising constraint of this project, and it is why roughly half the work reported here is about the label rather than the model.

1.3 Contributions

1. A **two-track labelling design** that keeps the auditable formula as a baseline and uses independent holistic judgment as the target, making the rubric–label agreement a measurable result instead of a tautology.

2. A documented **label failure and repair**. We show quantitatively that our first label measured the wrong construct, diagnose both causes, repair them, and re-judge all 900 pairs so that the two passes form a controlled comparison.
3. An explicit **null-is-not-zero missing-value policy**, motivated by a concrete case in which median imputation inverted the meaning of the strongest freshness feature.
4. A global ranking of 1,226 university websites with SHAP-based explanations, a per-university improvement report, leave-one-region-out fairness analysis, and a quantified statement of what the ranking cannot see.

2 Dataset and Audit

The raw dataset contains 1,230 university landing pages described by 69 attributes, organised into 11 blocks (B1 header/navigation, B2 rankings/recognition, B3 notices/updates, B4 events/media, B5 page content, B6 footer, B7 visual design, B8 service interaction, B9 technical performance, B10 SEO metadata, B11 accessibility). Every finding below was re-derived in code in notebook `01_audit` rather than assumed.

Table 1: Audit findings and the action taken. Nothing was silently corrected.

Finding	Scale	Action
Prestige columns almost entirely null	a10 99.35% null, a08 94.96% null	Dropped: unusable <i>and</i> a leakage risk
Exact duplicate column	load_time_s $\equiv$ a62	One dropped
Constant / empty columns	3 columns	Dropped
Duplicate domains	4 pairs (8 rows)	Merged to 1,226 rows; load time set to the pair median; mislabelled country fixed
Notice dates after the crawl date	285 of 759	Censored to the crawl date; flag retained
Notice flags contradict the dates	382 + 156 rows	Reconciled into an ordinal <code>notice_evidence</code> $\in \{0, 1, 2, 3\}$
Event flags contradict the counts	327 + 158 rows	Reconciled into <code>event_evidence</code>
Collector $\perp$ region is 1:1	6 collectors, 6 regions	Unresolvable ; reported as a limitation
Load time varies by region	median 8.07 s to 12.94 s	Region z -scored; every ranking produced with and without it
Country field mixes levels	10 bucket labels, 170 rows	Country ranking restricted to the 19 real countries with ≥ 20 universities

2.1 The missing-value policy: null is not zero, and not the median either

The most consequential decision in the pipeline

The median of `notice_recency_days` is **1 day**. Median-imputing it would tell every downstream consumer that the **469 universities with no dated notice anywhere on their site posted something yesterday** — handing the strongest freshness signal in the dataset to exactly the sites that earned it least. Absence was being read as average when it means *absent*.

Where absence has a known direction, the feature is filled at the worst defensible value instead, and the rule is written to `missing_value_policy.json`:

Feature	Rows filled	Fill	Meaning of absence
notice_recency_days	469	3650 d	no dated notice exists at all
a72_alt_text_pct	4	0%	no image carries a text alternative
a53_contrast_ratio	1	1.0:1	no readable text block was found

Companion `*_missing` indicator columns are computed *before* the fill, so “this was never measured” remains recoverable as a separate fact.

3 The Labelling Design

3.1 Track A — the transparent rubric (baseline, not target)

Track A is a rubric written and frozen *before* any score was computed. It applies:

- **Gates** that cap the achievable score regardless of everything else: no primary navigation caps at 45; no HTTPS caps at 60.
- **Six tiers**, heaviest first: can-I-do-the-task content; is-this-alive evidence; depth of service; accessibility and technical; polish; and self-promotion (ranking badges, trust seals, testimonials) at near-zero weight.
- **Non-linear response curves** that a correlation matrix cannot produce — an inverted-U on menu size (0–2 items is broken, 5–9 ideal, 15+ cluttered), diminishing returns on statistic counts, a plateau on contrast at 7:1 (WCAG AAA, since 21:1 is not better than 12:1), and half credit for undated notices.
- **Five personas** (domestic applicant, international applicant, accessibility specialist, researcher, literature-aligned), averaged into `trackA_consensus`, with the between-persona standard deviation retained as a per-university uncertainty.

Track A is a deterministic function of the features. A model predicting it would score $R^2 \approx 0.98$ and demonstrate nothing, so it is used strictly as **the baseline the learned model must beat**.

3.2 Track B — blind pairwise judgment (the target)

200 universities were sampled, stratified by region and across the full Track A range. Each was rendered as an **anonymised profile card** carrying no name, country, region, URL, collector or Track A score — blinding is what prevents prestige and rubric anchoring from leaking into the label. Cards were presented in 900 pairs: 500 random (for graph connectivity and wide gaps), 340 matched within ± 5 Track A points (the hard, informative cases), and 60 repeats re-presented later with the sides swapped. Left/right position was randomised.

Judgments were then fitted with the **Bradley–Terry** model,

$$P(i \text{ beats } j) = \sigma(\beta_i - \beta_j),$$

estimated by L_2 -regularised logistic regression on a ± 1 design matrix with no intercept, with bootstrap confidence intervals on each β_i . The resulting latent strength β is the supervised target `trackB_bt_score`.

4 The Relabelling: A Documented Repair

Our first labelling pass completed all 900 judgments and was then *discarded as a target*. We report it because the reason is a result, not an embarrassment.

4.1 Two defects

(a) **A corrupted feature reached the judge.** The profile cards were rendered from median-imputed features, so the 469 sites with no dated notice were described as “posted yesterday” (Section 2.1).

(b) **The cards led the judge to the wrong criterion.** 37.6% of the first pass’s free-text reasons cited *alt-text percentage*. Alt text is invisible to a sighted visitor and was on the card only because it happened to be a numeric attribute. The label was measuring machine-readable accessibility metadata rather than the quality of the website as a website.

4.2 The repair

Cards were re-rendered, grouped by *what a prospective student is trying to do* rather than by attribute block — APPLY, STUDY, ALIVE, NAV, WORKS — with accessibility metadata and marketing badges demoted to the last two lines. The judging question was restated concretely:

You are a prospective student deciding where to apply. You have never heard of either university. Based on the website alone, which one would you feel more confident applying to? Weighing heavily: can I find the programmes, the requirements, the deadlines and the fees; can I reach a human; is the institution visibly active; can I find my way around; can I actually read the page. Weighing lightly: badges, galleries, videos, social-media links.

The same 200 universities and the same 900 pairs were re-judged, which makes v1 → v2 a controlled comparison rather than a new experiment.

Table 2: What the judge’s stated reasons cite, before and after the repair (% of 900 reasons).

Reason cites	Pass 1	Pass 2	Reading
alt-text / screen-reader	37.6%	0.0%	the invisible attribute is gone
admission-task content	38.1%	73.1%	nearly doubled; now dominant
freshness	7.9%	26.3%	tripled, once measured honestly
navigation	35.1%	41.7%	broadly stable
Winners that changed: 16.4% (148 of 900)			

Why 16.4% is the right number

Had the rewrite changed nothing we would expect ~2% (rater noise). Had the judgments been arbitrary we would expect ~45% (near-random). We observe 16.4%: obvious pairs held, close pairs moved. That is the signature of a genuine change of criterion.

4.3 Effect on the model

The repair propagates all the way to what the model learns. Under the first label, alt-text coverage was among the strongest permutation-importance features; under the applicant-lens label it does not reach the top fifteen, which is now headed by content completeness, department links, menu size, admissions policy, contrast and notice recency. Both the self-consistency of the label and the accuracy of the model improved (Table 3).

Table 3: Effect of the relabel on label reliability and held-out model accuracy.

	Pass 1 label	Pass 2 label
Self-consistency (60 swapped repeats)	86.7%	96.7%
Spearman(Track A, Track B)	0.790	0.814
Held-out test Spearman ρ ($n = 40$)	0.885	0.922
Held-out test R^2	0.752	0.818
Held-out test MAE (points)	8.92	9.02

5 Label Validation

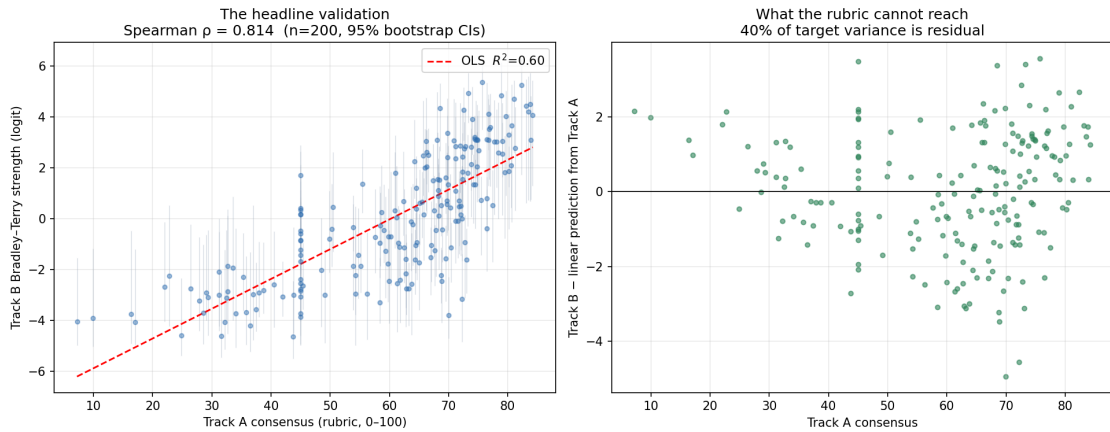
Before the label was used as a target, five checks had to pass.

Table 4: Validation of the 900 judgments and the Bradley–Terry fit.

Check	Value	Interpretation
Judgments collected	900	840 unique pairs + 60 swapped repeats
Self-consistency	96.7%	the rater is stable; CI [0.885, 0.996]
Position bias	48.2% left	binomial $p = 0.30$; no side preference
Bradley–Terry fit (held-out comparisons)	82.7%	the latent score genuinely explains the judgments
Spearman(Track A, Track B)	0.814	correlated — both measure quality
Variance <i>not</i> explained by Track A	40%	... yet the label is not that formula in disguise
Rubric predicts <i>wide-gap</i> pairs	88.0%	convergent validity when the answer is obvious
Rubric predicts <i>close</i> pairs	62.4%	the formula collapses towards chance

Read the last two rows together

When two websites differ obviously, a weighted formula and a holistic judgment agree (88%). When they are close, the formula is barely better than a coin flip (62%) — it cannot separate two sites with similar feature counts but different execution. **That gap is precisely the signal the model is asked to learn**, and it is why this is a genuine supervised learning problem rather than curve-fitting to our own arithmetic.

**Figure 1:** Left: the learned label against the rule-based rubric. Right: where the rubric breaks down — accuracy on wide-gap pairs versus close pairs.

6 Modelling

6.1 Protocol

The target is the raw Bradley–Terry logit, which is the statistically correct scale — unbounded and centred. The 0–100 presentation score is a monotone rescaling used for reading only and is never the training target. Of the 200 labelled universities, **160 train / 40 test**, stratified on score band. **The test set is touched exactly once**, in Section 6.3; no feature, hyper-parameter or threshold was chosen using it. Model selection uses nested cross-validation (outer repeated 5-fold $\times$ 5 seeds, inner 3-fold tuning), with all imputation, scaling and selection fitted inside the training fold.

Twenty-two columns are excluded from the design matrix: identifiers, crawl metadata, the two prestige *value* columns (unusable and leaky), `load_time_s` and `member` (confounded with region), and features superseded by engineered replacements. Prestige *display* flags — does the site show a QS badge or a national rank claim — are retained, because “the site advertises a badge” is a property of the website. 78 features remain.

6.2 Model comparison

Table 5: Nested cross-validation on the 200 labelled universities. LORO = leave-one-region-out. The row that matters is **Track A composite**: if a transparent weighted sum matched the learned models, the machine learning would contribute nothing and we would report that.

Model	CV ρ	LORO ρ	MAE	R^2	Verdict
LightGBM	0.881	0.835	0.99	0.761	beats the rubric
LightGBM + monotone	0.877	0.840	1.00	0.754	beats the rubric
LightGBM + label weights	0.873	0.844	1.00	0.752	selected
RandomForest	0.861	0.825	1.08	0.705	beats the rubric
Ridge	0.855	0.830	1.13	0.687	beats the rubric
Track A composite (baseline)	0.810	0.774	1.32	0.590	—
B5 content block only	0.752	0.693	1.39	0.521	loses to the rubric
B9 technical block only	0.199	0.100	2.14	0.025	loses to the rubric
Mean predictor	0.000	0.000	2.15	−0.001	loses to the rubric

The selected model is **LightGBM with label weights** $w_i \propto 1/\text{SE}(\beta_i)$, so universities whose latent score was estimated confidently count for more. It is not the best on raw cross-validation — plain LightGBM is, at $\rho = 0.881$ — but it is the best under **leave-one-region-out**, which is the harder and more honest test of whether the model learned quality rather than geography. All learned models beat the rubric with $p < 0.001$ on a paired test across the 25 fold-repetitions.

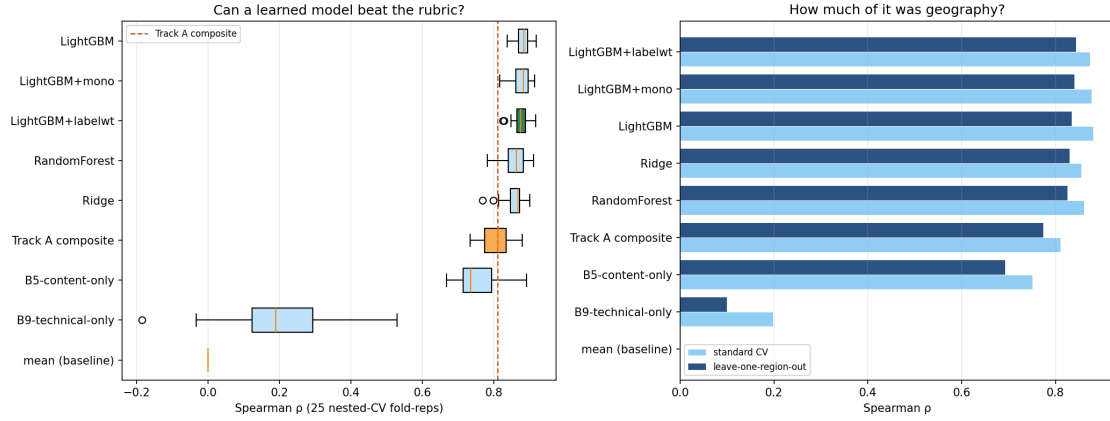


Figure 2: Model comparison. The dashed line is the transparent rubric baseline.

6.3 Held-out test result

Held-out test set, $n = 40$, used once	
Spearman ρ (rank agreement)	0.922
Kendall τ	0.736
MAE (points on 0–100)	9.02
RMSE	11.44
R^2	0.818
Predictions within ± 10 points	62%
Predictions within ± 15 points	78%

7 Results

7.1 The ranking

All 1,226 universities are scored; the 1,026 without an expert label are pure inference and were never used in fitting. Ties are broken transparently: predicted score, then content completeness, then accessibility completeness, then `uni_id`.

Global top 8			Bangladesh top 6 (of 22)			
#	University	Score	#	University	Global	Score
1	Murdoch University (AU)	92.2	1	KUET	31	85.9
2	University of Otago (NZ)	92.0	2	RUET	66	81.2
3	RWTH Aachen University (DE)	91.8	3	CUET	86	79.5
4	Southern Cross University (AU)	91.7	4	North South Univ.	137	76.7
5	University of Tartu	90.8	5	Khulna University	142	76.4
6	Polytechnic Univ. of Milan	90.5	6	Univ. of Professionals	161	75.7
7	University of Houston (US)	90.2				
8	University of Porto (PT)	90.0				

On KUET's position — reported

KUET scores **85.9 / 100**, **grade A+**, **global rank 31**, **first in Bangladesh**. This is higher than a casual visitor might expect, and we did not adjust it. The reason is visible in the data: on the attributes this dataset actually measures, KUET's landing page ticks nearly every box — programmes listed, departments, faculty, library and research pages present, a notice board posted within the last month, admission notices, scholarships, contact route, HTTPS, no broken links, contrast 13.3:1. What the dataset *cannot* see is visual design, layout density or whether the page looks modern (Section 8). A site that satisfies every content requirement will rank highly here even if a human would find it dated. We regard reporting this openly as more valuable than hand-tuning the score to match intuition.

7.2 What drives the score

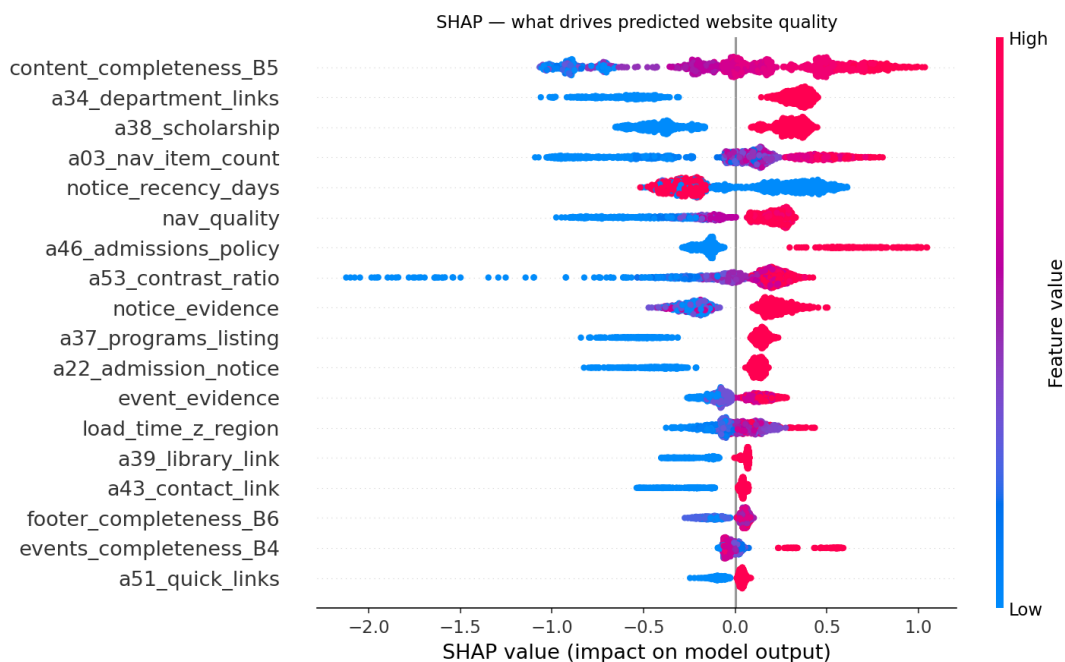


Figure 3: SHAP values on held-out folds. The score is driven by whether the site serves a visitor trying to apply: content completeness, department links, menu size, admissions policy, contrast and notice recency.

7.3 Declared weights versus realised influence

This is the project's substantive claim about rubric design. Track A *declares* a weight for each block; SHAP *measures* the share of ranking variation each block actually drives.

Table 6: Declared rubric weight versus realised SHAP influence, by block.

Block	Declared %	Realised %	Ratio
B10 SEO metadata	3.6	0.7	0.18
B8 service interaction	1.8	0.3	0.19
B11 accessibility	6.4	1.3	0.20
B6 footer	5.4	3.8	0.70
B4 events / media	8.9	6.4	0.72
B9 technical performance	6.2	5.4	0.86
B2 rankings / badges	2.1	2.4	1.12
B5 page content	37.9	43.5	1.15
B1 header / navigation	11.5	13.6	1.19
B3 notices / updates	12.7	16.7	1.32
B7 visual design	3.4	6.0	1.74

Accessibility is the sharpest case: declared at 6.4%, it drives 1.3% — five times less than assigned. Under our *first* label it appeared far more influential, for the reason diagnosed in Section 4. Conversely, visual design and notice freshness both drive more than they were given. This reproduces, on a second dataset and with a different method, the effect Rashida et al. report: they allocated 40% of their score to performance, which then drove 2.9% of their ranking variance, because every university scores alike on it. Hand-set weights misallocate importance across quality dimensions, and only measuring realised influence reveals it.

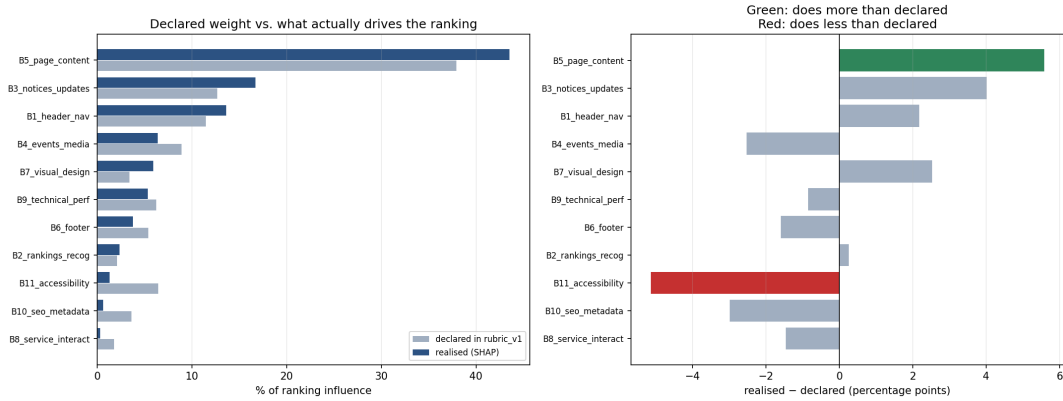


Figure 4: Declared rubric weight against realised SHAP influence per block.

7.4 Fairness and the collector confound

Each of the six data collectors covered exactly one region, so **collector and geography are perfectly confounded and cannot be separated with this data**. We therefore report model *accuracy* by region rather than claiming regional score differences are real:

Region	N. America	S. Asia+Oc.	W./N. Europe	E./S. Europe	E. Asia	LatAm/Africa
LORO ρ	0.811	0.826	0.852	0.854	0.854	0.869

Accuracy is stable across all six (0.81–0.87), so the model is not a regional lookup table. Load time is the feature most contaminated by the confound, so every ranking was produced twice, with and without `load_time_z_region`: the two orderings correlate at $\rho = 0.992$, with a median displacement of 23 places and 264 universities moving more than 50 places. Load time therefore matters at the margin but does not determine the ranking.

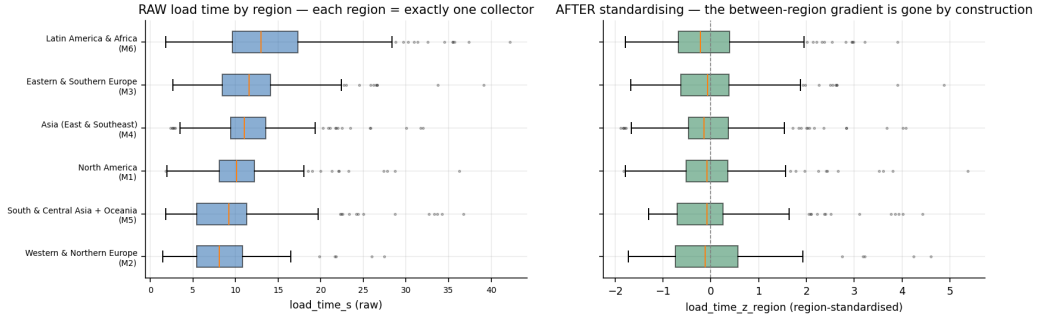


Figure 5: Load time by collector: the confound that motivates region standardisation.

8 Independent Cross-Check in Weka

The same 160/40 split was exported to ARFF and re-run with off-the-shelf Weka schemes, as an independent check that the label is learnable and not an artefact of one library.

Weka scheme	Correlation	MAE	RMSE
ZeroR (baseline)	0.00	22.8	26.8
LinearRegression	0.82	12.1	15.4
REPTree	0.71	15.7	19.6
RandomForest (100)	0.88	11.3	14.2
<i>LightGBM (this project, Python)</i>	<i>0.92</i>	<i>9.0</i>	<i>11.4</i>

Weka reproduces the ranking at correlation 0.88 with a plain random forest. A six-class grade classifier reaches 42% exact accuracy but 82% within one grade — the residual error is boundary noise from cutting a continuous score into bands, which is precisely why the project treats the task as regression. Full click-by-click instructions are in `WEKA_GUIDE.md`.

9 Limitations

We state these plainly because several of them bound how the ranking should be read.

1. **The dataset cannot see design.** Every feature is a presence flag, a count or a measurement. Whether a page is cluttered, dated-looking or unpleasant to use is not in the 78 columns, so it cannot be in the score. This is the largest gap between this ranking and a real user’s impression, and closing it would require screenshots and human raters rather than more modelling.
2. **2.0% of rows are extraction failures, and they contaminate the bottom of the ranking.** 25 universities have no navigation, no programmes list, no departments page and no contact link recorded — almost certainly JavaScript-rendered pages the crawler could not read, not genuinely empty websites. 21 of them fall in the bottom 100. The *top* of the ranking is unaffected; the bottom should not be quoted without this caveat.
3. **The labels are expert-style judgments made from extracted attribute profiles,** not ratings of live websites by a panel of humans, and must be described that way.
4. **One rater.** Self-consistency is measured (96.7% on swapped repeats); inter-rater agreement cannot be, because there is only one rater.
5. **No external validation** against QS, Webometrics or a student survey was performed, so no claim of agreement with real user perception is made.
6. **Collector and region are perfectly confounded** and this dataset cannot separate them. The regional ranking is the safer artefact to quote.

7. **200 labelled of 1,226.** The other 1,026 scores are model inference, flagged by the `has_expert_label` column.
8. **a66_broken_links has no denominator**, so 12 broken links on a 20-link page and on a 2000-link page are recorded identically.

10 Conclusion

We set out to rank university websites by quality and found that the hard part was defining quality without defining it circularly. Separating the transparent rubric (baseline) from blind pairwise judgment (target) turns rubric–label agreement into a measurable result: they correlate at $\rho = 0.814$, but the rubric predicts close pairs at only 62.4%, and it is that residual 40% of variance the model learns. On a test set used exactly once, the selected LightGBM reaches $\rho = 0.922$ and $R^2 = 0.818$, comfortably beating the rubric at $\rho = 0.810$, and holds $\rho = 0.844$ under leave-one-region-out.

The finding we would most want carried forward is methodological. Our first label looked perfectly healthy by its own statistics — 86.7% self-consistent, no position bias, correlated with the rubric at 0.790 — and was nonetheless measuring the wrong thing, because the evidence we showed the judge foregrounded an attribute no visitor can perceive and a missing-value default that inverted the meaning of freshness. No summary statistic on the label would have caught it; reading the judge’s stated reasons did. Labels deserve the same auditing we routinely give models.

A Reproducing this work

Path	Contents
<code>data/university_websites_labeled.csv</code>	all 1,226 universities, scored and ranked
<code>data/train.csv</code> , <code>data/test.csv</code>	the 160/40 expert-labelled split
<code>data/label_provenance/</code>	both judgment passes, pairs, cards, fit statistics
<code>data/weka/</code>	the same splits as ARFF
<code>notebook/University_Website_Quality_Ranking.ipynb</code>	the full pipeline, executed
<code>model/final_model.joblib</code>	the deployed model
<code>WEKA_GUIDE.md</code>	step-by-step Weka instructions

The research pipeline that produced these artefacts is the nine notebooks `01_audit` through `09_deliverables` in the parent project directory, which run in order from the raw CSV with a fixed seed.

Environment. Python 3.13, pandas 2.3.3, numpy 2.4.4, scikit-learn 1.8.0, lightgbm 4.6.0, shap 0.51.0.